

Ethan Tapia

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EDUCATION

Chapman University

Orange, CA

Bachelor of Science in Computer Science, Themed Inquiry in Design and Fabrication

May 2026

Related Coursework: Systems Programming, Operating Systems, Database Management, Data Structures and Algorithms,

Algorithm Analysis, Digital Logic Design, Computer Architecture, Data Networks and Comm.'s, Theater Technology, Lighting Design, 3D Printing and Design.

TECHNICAL SKILLS

Languages: C#, C++, C, Python, Java, MATLAB, R, SQL

Industrial Controls: Beckhoff TwinCAT, EtherCAT, PLC Programming, Elmo Servo Drives, Embedded Control Systems, H.I.L. Testbed Development

Frameworks & Web: .NET, Next.js

Libraries: OpenCV, DLTdv, NumPy, pandas, Matplotlib

Developer Tools: Git, GitLab, Bitbucket, Bamboo, Jira, Confluence, Docker, VSCode, MATLAB Simulink, AutoCAD, SolidWorks, Vectorworks, xLights

Practices: System Testing, Regression Testing, Root Cause Analysis, Real-time Systems, Digital Logic Design, State Machine Design

PROFESSIONAL EXPERIENCE

Software Engineering Intern

June 2026 – Present

The Walt Disney Company

Orlando, FL

- Developed and deployed C#/.NET features applying object-oriented design to real-time attraction control systems, validating changes on Beckhoff TwinCAT/EtherCAT environments interfaced with Elmo servo drives
- Migrated a decade-old ride-control diagnostics tool integrated with Maximo logging onto a React/Node.js stack, closing out a legacy technical-debt ticket open for nearly two years and improving fault-identification tooling across ride control systems
- Partnered with Walt Disney Imagineering (WDI) to test and commission control, hardware, and animatronic systems on live and in-development attractions, diagnosing and resolving software and hardware issues

Systems & Software Engineering Intern

June 2025 – Oct. 2025

Diality Inc.

Irvine, CA

- Collaborated with cross-functional teams to perform system-level testing and develop hardware-in-the-loop testbed environments for hemodialysis device validation, ensuring compliance and subsystem integration
- Conducted root cause analysis of system failures using Python, C++, and cloud log data to improve robustness
- Developed internal automation tools in C++ that streamlined testing workflows and improved efficiency by 30%, supporting quality assurance across legacy and next-generation systems

Engineering Research Assistant

July 2023 – Feb. 2025

Chapman University

Orange, CA

- Designed and optimized cross-functional engineering prototypes using C++, MATLAB Simulink, Python, and R for system modeling and real-time control simulations
- Applied advanced testing and data-driven feedback loops to evaluate performance, securing a \$4,000 research grant

Property Manager

June 2020 – March 2026

Fairgrove Property Management

Irvine, CA

- Coordinated cross-functional teams including maintenance technicians, contractors, and vendors to deliver property improvement projects; modernized operations with AppFolio, boosting productivity by 20% while sustaining 98% occupancy for 5 consecutive years

PROJECTS

Heritage Fest | C++, Python, AutoCAD, Embedded Controllers

- Led technical development and creative design teams for large-scale decorative light and projection-mapped entertainment shows at community events serving a residential community
- Programmed embedded logic controllers for real-time audiovisual synchronization across lighting, projection mapping, and sound systems, implementing state machines for multi-zone coordination
- Designed and fabricated custom control hardware interfaces integrating DMX lighting protocols with projection systems

F.I.S.H.Bot | C++, Python, MATLAB, Simulink, R, AutoCAD

- Developed an autonomous swimming robot with biomimetic control systems, implementing digital logic for motor control and sensor integration using C++ with MATLAB Simulink for real-time simulation and control parameter tuning
- Built experimental testbeds with particle image velocimetry (PIV) to analyze water flow dynamics and optimize hydrodynamic efficiency through iterative hardware-software integration

Corgicade | AutoCAD

- Designed and built a modular arcade cabinet for Chapman University using AutoCAD for design, managing CNC fabrication and cross-team assembly coordination to translate design into final build